

Yi Xia (Sean)

Applied AI Engineer · Computer Vision · End-to-end ML Deployment

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Applied AI engineer who ships **computer-vision systems to production**. Four years at **TOHU MEDIA**, a Māori-owned media-production company, designing and deploying object-detection, OCR and action-recognition pipelines end-to-end — problem framing, dataset spec, PyTorch training, **Docker-packaged inference on cloud GPU servers**, and iteration with end users on what "useful output" looks like inside a real workflow. Concurrently completing a PhD at **Auckland University of Technology** on deep learning for human activity recognition in video; two first-author Q1 journal papers (**Pattern Recognition** and **Computer Science Review**, 2026), seven peer-reviewed papers, 70+ citations. Open to **Applied AI, Machine Learning, Computer Vision or Data Science** roles with a New Zealand team that turns research into reliable production systems.

PRODUCTION Object-detection, OCR & action-recognition systems running in a live media-production pipeline — four years.

END-TO-END Dataset spec · PyTorch training · Docker inference on cloud GPUs · iteration with end users.

DEPTH PhD candidate (AUT) · first-author in two Q1 journals · seven peer-reviewed papers, 70+ citations.

TECHNICAL SKILLS

Production & Deployment

Docker · containerised model inference on cloud GPU servers · full ML lifecycle ownership · latency ↔ accuracy trade-offs

Computer Vision

Object detection (incl. YOLO family) · object tracking · OCR · action recognition · semantic / instance segmentation · classification · video analytics

Deep Learning

PyTorch (primary) · TensorFlow / Keras · OpenCV · scikit-learn · NumPy · pandas · Hugging Face

Programming

Python (expert) · Java

Generative AI & LLM

LLM integration · agentic workflows · agent skills & tool use · Claude / GPT APIs · prompt engineering

Tooling

Git / GitHub · Linux · CUDA · Jupyter

DOMAIN EXPOSURE

- Media / film production** — current TOHU MEDIA work: object detection, OCR & action recognition on long-form video
- Agriculture / horticulture** — published research on kiwifruit detection & tracking from orchard video (50+ citations)

- Activity recognition** — PhD line on skeleton-based human action understanding; applications across healthcare, sports, safety and elder-care monitoring

EDUCATION

PhD, Computer & Information Sciences

Auckland University of Technology
Oct 2023 – Oct 2026 (expected)

Deep learning for skeleton-based human activity recognition. **AUT PhD Full Scholarship**.

Master of Computer & Information Sciences

Auckland University of Technology
2021 – 2023 · Specialisation: Computer Vision

BEng, Logistics Engineering (Software)

Dalian Neusoft University of Information, China
2011 – 2015

SELECTED EXPERIENCE

Artificial Intelligence Developer at TOHU MEDIA

Auckland · Feb 2022 – present

Full-time since June 2025; previously part-time alongside MSc and PhD

- **Own the company's AI capability end-to-end** at a Māori-owned media-production company — designed and shipped **three production AI systems: object detection, OCR and action recognition**, replacing previously manual transcription work on client video footage with automated pipelines (specific deliverables under NDA).
- Built CV systems that **work on real production footage**, not just clean benchmark data — handled variable shot conditions, dense overlay text in non-standard layouts, archival material with poor contrast, and long-form takes that exceed single-workstation GPU memory. Selected model architecture, training regime and post-processing accordingly.
- Took every system through the **full ML lifecycle**: dataset scope and labelling spec, PyTorch training and evaluation, **Docker-packaged inference deployed to the company's cloud GPU servers**, and iteration with editors and the creative director on what "useful output" means inside an actual editing workflow. Every model is trained and served on infrastructure the company controls — **full data sovereignty** over client footage.
- Made independent calls on architecture, **latency / accuracy trade-offs** and scope reduction under deadline — four years part-time alongside the MSc and PhD, now full-time ownership of the AI work.

PhD Researcher at Institute of Robotics & Vision, AUT

Oct 2023 – Oct 2026

Supervisors: Dr Raymond Lutui (primary) · Assoc. Prof. Sira Yongchareon

- Designed a **frequency-aware spatio-temporal graph network** for skeleton-based human activity recognition; outperforms prior state-of-the-art on standard benchmarks — accepted at *Pattern Recognition* (Q1 journal, IF = 7.6) in 2026.
- First-authored a survey on generalisation in skeleton-based HAR in **Computer Science Review** (Q1 review journal, 2026), plus a peer-reviewed paper at **ICONIP 2024** on dynamic self-attention gating between spatial and temporal streams.
- **Full ownership of first-author research** — idea, model design, GPU training, ablations, paper writing and reviewer rebuttal.
- Mentor master's students on PyTorch, experimental hygiene and academic writing.

Research Assistant at School of Engineering, Computer & Mathematical Sciences, AUT

Nov 2022 – Jul 2023

- Funded research assistant on the **kiwifruit detection & counting** programme — co-built KiwiDetector (improved YOLOv7) and KiwiTracker for real-time orchard video. The lead paper now has **50+ citations**.
- Co-authored three peer-reviewed papers (IVCNZ 2022, SAI 2023, PSIVT 2023) during the assistantship.

Teaching Assistant at AUT

2022 – present

- TA on multiple undergraduate computer-science courses (programming, computer vision, AI). Run labs, mark assignments, hold consultation hours and mentor on final-year projects.

- Handled customer returns and exchanges, first-line hardware diagnostics and repairs, and custom-order PC builds in a busy retail service environment — sharpened plain-English technical communication with non-specialist users.

Co-Founder · Junior Lecturer at *Baiyin Pinuo Training School · Baiyin Vocational & Technology College* China · 2015 – 2021

- Junior Lecturer (2015 – 2019):** taught Computer Fundamentals, Java, IoT applications and RFID technologies to ~120 students per year.
- Co-Founder (2019 – 2021):** co-founded a private training school; built curriculum, ran operations and finance; divested in 2021 to relocate to New Zealand for postgraduate study — the school remains in operation under my co-founder.

SELECTED PUBLICATIONS

7 peer-reviewed papers · 70+ citations

Y. Xia, S. Yongchareon, R. Lutui, Q. Z. Sheng (2026). "Beyond Benchmark Accuracy: A Generalization-Centered Analysis of Deep Learning for Skeleton-Based Human Activity Recognition." *Computer Science Review*. **Q1 review**
doi.org/10.1016/j.cosrev.2026.101030

Y. Xia, S. Yongchareon, R. Lutui, Q. Z. Sheng (2026). "Frequency-Aware Spatio-Temporal Topology Learning for Skeleton-Based Human Activity Recognition." *Pattern Recognition*. **Q1 journal**
doi.org/10.1016/j.patcog.2026.113146

Y. Xia, M. Nguyen, W. Q. Yan (2022). "A Real-Time Kiwifruit Detection Based on Improved YOLOv7." *IVCNZ 2022*, Springer LNCS. **50 citations**
doi.org/10.1007/978-3-031-25825-1_4

Y. Xia, S. Yongchareon, R. Lutui, Q. Z. Sheng (2025). "Dynamic Self-Attention Gated Spatial-Temporal GCN for Skeleton-Based HAR." *ICONIP 2024*, Springer LNCS.
doi.org/10.1007/978-981-96-6588-4_14

+ three further peer-reviewed papers on kiwifruit detection & tracking — full list on [Google Scholar](#).

REFERENCES

Available on request. Doctoral supervisors (Dr Raymond Lutui, AUT; Assoc. Prof. Sira Yongchareon, AUT) and an industry referee from TOHU MEDIA can be provided.